Teaching Philosophy

My career path and world view has been greatly shaped by engaging classes and enthusiastic teachers. My goal as a teacher is to provide students with the same sense of belonging and awe that inspired me to pursue my own academic career—to excite them towards their own path of discovery and self-actualization. I aim to do so through communicating the importance of ecological and evolutionary principles for understanding our changing world, while focusing on foundational and transferable skills which are fundamental to both science and our society.

Teaching through Mentorship and in the Classroom

Mentorship provides a platform to create, transform, and share knowledge that is fundamental to my research. I have mentored six undergraduate students on research projects, two of which have led to papers published (Kreiner 2021, *Evolution*) or in prep. Through my roles in the UBC Botany EDI committee and the Biology Undergraduate Diversity in Research Program, I work through mentorship to contribute insights on career and academic development to undergraduate students from historically underrepresented groups. Each of my supervisory experiences has reinforced the importance of building on individuals unique interests, reinforcing their strengths, and supporting their career goals. To help my students push their intellectual boundaries, I emphasize three foundational skills—scientific literacy, quantitative inference, and effective communication. Each student learn hows to read and discuss the primary literature, to inspire their own hypothesis development and experimental designs. Students build an intuition for statistics and problem solving by analyzing and visualizing their data in command line and R, while learning how to synthesize inferences from their results. Finally, we work towards producing a clear scientific dialogue through iterations of feedback on proposals, final papers, and oral presentations, with the goal of engaging diverse audiences with our discoveries.

I have found that these foundational skills are just as relevant in course settings, and I will continue focus on them while fostering an inclusive classroom environment built on universal designs for learning. Comprehension of scientific concepts and quantitative inference will be emphasized and assessed, rather than rote memorization. Equally as important to my teaching is creating a space where students are actively engaged, excited to learn, and can safely grow. I have done this in the classroom by creating opportunities for small group discussions, peer review, interactions with me (the instructor), and by soliciting feedback throughout the course to give students agency to shape their own learning experience. I will work to demystify the hidden curriculum for my students, setting them up for success in my courses by clearly outlining learning objectives, pathways to graduate school, and connections to a diversity of careers. In both my lab and my courses, I will develop a code of conduct that requires respectful treatment of all axes of diversity and sets accountable goals for professional and personal growth.

Teaching Experience & Future Courses

Throughout my Ph.D., I instructed lab sections for lower-level evolution and ecology courses (BIO120, BIO220), and was a teaching assistant for upper-level undergraduate evolution and graduate programming courses (EEB318, EEB324, EEB1450) for which I also guest lectured. These environments prepared me for delivering complex material to large audiences. I facilitated student-led, discussion-based tutorials for Evolutionary Ecology, where we focused on critical evaluation of primary literature and synthesis of key outstanding questions in the field. This type of activity mimics the critical thinking and communication skills developed in small discussion groups I have organized (e.g. Plant Evol. Gen., Student Journal Club, Evolution Book Club), and which were central to my own professional growth. I will carry these exercises into my undergraduate and graduate teaching, which could range from broad courses such as evolutionary ecology, evolutionary genetics, or population genetics to more specific topics detailed below.

As the Bioinformatics Research Fellow at UBC, I have had the pleasure of developing and instructing the *Bioinformatics for Evolutionary Biology lab-based graduate and upper-level undergraduate course.* This workshop brings active learning to the forefront with student time spent predominantly in computational labs. By working from the basics of command line and processing of raw genomic data to interpreting results in a hypothesis-driven framework, we aim to provide an accessible introduction to programming for evolutionary biology. To supplement hands-on labs, I developed lecture material from genome assembly to effective data visualization of genomic data, while highlighting the latest breakthroughs in genomics. We wanted to provide student the opportunity to work with a variety of data types but also consider the pros and cons of different sequencing designs for hypothesis testing. To do so, we used evolutionary computer simulations (i.e. SLIM) to generate genetically structured populations adapted to a climate gradient and sampled them with different types of genomic technologies, varying in coverage, depth, read length, and error rate. We have received extremely positive feedback on this course, with a 95% average student satisfaction rate, with one student stating “[it was] one of the best courses I’ve taken”.

Exposing students to the ever-changing nature of science and discovery is central to my teaching philosophy, not only with historical context but real-world current-day examples. Along these lines, I am interested in developing *a lecture and discussion-based course on Genomics & Society*. Working in genomics, I am acutely aware of the public's lack of understanding of genetics and the resulting mistrust of agricultural technological advances and misinterpretation of human genetic risk factors. The curriculum for this course would thus contrast the history of and motivation for genomic technological advances, global applications, and implications for legal and bioethical issues. Topics would include genetically modified technologies for plant breeding, global farming, and legislation; tracking the spread of virus and resistance variants with genomics (e.g. *nextstrain.org*); gene drive and pest control; personal genomics (e.g. 23andme) and the study and interpretation of human disease, ancestry, and race; and the eugenic history of population genetics. A key project would be to write an evidence-based blog for the public that discusses one such controversial topic or the implications of a new genomic biotechnology.

Interwoven with my interest and expertise in bioinformatics and genomics, is bringing plants to the forefront of the study of evolution and ecology. I would be thrilled to develop and teach *a lecture and lab-based course on The Plant Genome*. The curriculum would explore the consequences of plant diversity and human impacts on genome structure and evolution drawing on case-study and inquiry-based learning techniques. Key topics could include the impact of transitions in mating system and ploidy on patterns of genetic diversity; genomic signatures of sexual dimorphism and sex chromosome evolution; variation in recombination rate, selection, and introgression across the genome; and the impact of humans on plant genome evolution (e.g. domestication, landscape fragmentation, human-mediated dispersal). Lecture-based material would heavily rely on case studies from the primary literature with accompanying written critiques as assignments, and the final project would develop a thesis on a key outstanding question in the field. With appropriate support, the lab component could sequence and analyze the genome of an at-risk or conceptually interesting plant species, at the choice of the students.